

Multiple Choice (10 marks)

1. Write the ratio of 96 runners to 216 swimmers in simplest form.
 - (a) 96 over 216
 - (b) 16 over 36
 - (c) 4 over 9
 - (d) 9 over 4
2. Jess scored 18 points during her last basketball game. Each basket she made was worth 2 points. How many baskets did she make?
 - (a) 20
 - (b) 16
 - (c) 9
 - (d) 8
3. Find the number that makes the statement true: $48 \text{ m} = \text{Box mm}$.
 - (a) 48,000
 - (b) 480
 - (c) 4,800
 - (d) 5
4. Which number written in exponential notation is equivalent to 343?
 - (a) 3^7
 - (b) 7^3
 - (c) 7^{49}
 - (d) 294^{49}
5. Emil wants to find out the most popular football team at a game between the home team and the visiting team. Which of the following methods will give him the most accurate results?
 - (a) surveying the cheerleaders for the home team
 - (b) surveying people wearing hats for the visiting team
 - (c) surveying a group of people standing in line for tickets
 - (d) surveying people who do not live in the home team's city

True / False (10 marks)

Answer True or False

6. True or False: The simplified form of $\frac{2+2i}{-3+4i}$ is $\frac{2}{25} + \frac{-14}{25}i$.
7. True or False: The expression $(2 \times 4) \times 6$ is equivalent to 8×6 .
8. True or False: The correct answer to 'Find x , given that

$$\frac{\sqrt{x}}{x\sqrt{3} + \sqrt{2}} = \frac{1}{2x\sqrt{6} + 4}.$$

' is $\frac{1}{2}$ '.

9. True or False: The object reverses direction two times in the interval from $t = 0$ to $t = 10$.
10. True or False: The product of 32 and 67 is 2,144.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A child has a set of 96 distinct blocks. Each block is one of 2 materials (plastic, wood), 3 sizes (small, medium, large), 4 colors (blue, green, red, yellow), and 4 shapes (circle, hexagon, square, triangle). How many blocks in the set differ from the 'plastic medium red circle' in exactly 2 ways? (The 'wood medium red square' is such a block) (A) 29 (B) 39 (C) 48 (D) 56 (E) 62
12. Discuss the relationship between the number of pears and apples in Simon's fruit salad. How would you express this relationship as a ratio, and what does it signify in this context?

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